

Numerical Implementation and Reproducibility

A Nonlocal Attraction–Repulsion Transport Equation

This document describes the supplied `matlabcode.m` script, in the order of the numerical examples in the manuscript: one dimension followed by two dimensions. Each stationary calculation below starts with the mathematical problem, explains the representation being used, and then describes its discretization and output. The numerical target is the stationary measure selected by the manuscript’s characterization, rather than the background measure itself.

How the four stationary calculations fit together. In one dimension, Section 2.1 solves the finite-interval Riesz equation and selects the support by the mass condition. Section 2.2 describes the Green/Boggio reconstruction of Proposition 7.1 and explains the exterior-extension implementation used to compute its comparison curve. In two dimensions, Section 3.1 solves the radial force-balance equation for ring masses. Section 3.2 constructs a second profile through the exterior disk Green formula, with a scalar coefficient calibrated in the supplied implementation. In each dimension, the subsequent particle evolution supplies a third distribution to compare with these stationary computations.

1 Running the code

Place `matlabcode.m` in a writable MATLAB working directory and run

```
matlabcode
```

The script clears workspace variables, closes open figures, and clears the command window. It contains its own local functions and needs no external input data. It uses `hypergeom` for the Green-kernel evaluation; a MATLAB installation providing this function is required (normally through Symbolic Math Toolbox). It uses `lsqnonneg` when available and supplies a projected-gradient fallback otherwise. It also calls `fzero`, `fminbnd`, `besselj`, `quantile`, and plotting functions including `xline`, `yline`, and `sgtitle`. A recent MATLAB installation supporting local functions in scripts and these plotting functions is appropriate. The source does not record the MATLAB release or toolbox versions used for the original figures.

The generated files are

- `fig_1d_a_greater_r_phi_compare.pdf` and its PNG counterpart;
- `fig_2d_ar_phi_compare.pdf` and its PNG counterpart.

Export uses `exportgraphics`, with vector content for PDF and 220 dpi for PNG. If export fails, the script uses `print`. Existing outputs with the same names may be overwritten. Console output includes masses, residuals, CDF discrepancies, and the one-dimensional final diameter. This document records the algorithm by source inspection; it does not report newly executed MATLAB results.

2 One-dimensional experiment

2.1 Recovering the one-dimensional stationary density and its support

Original problem. The target is a nonnegative probability density supported on an unknown symmetric interval $I = [-L, L]$, with zero force on its interior:

$$B(x) = V'(x) - (\psi_r * \phi)'(x) = 0, \quad V = \psi_a * \omega_m, \quad \int_{-L}^L \phi(y) dy = 1.$$

This is the interval stationary-state construction in the manuscript's one-dimensional example. The force balance and mass constraint are stated above, and their integral-equation formulation is derived below. Both the profile and the support half-width are unknown.

Why solve a Riesz equation? Differentiating the force identity inside the interval gives

$$r(1+r) \int_{-L}^L |x-y|^{r-1} \phi(y) dy = G_a(x), \quad G_a = V''.$$

For fixed L this is a linear integral equation for the density. Its differentiated form determines the force only up to a constant. For the even background and even interval solution considered here, the force vanishes at $x = 0$ by symmetry; this supplies the one-point force condition in the proposition. The mass-one condition then selects L . This explains the nested algorithm: solve a linear density problem for each trial interval, then solve one scalar equation for the interval size. Normalizing an arbitrary trial density alone would generally destroy its force balance, which is why the support is selected through the mass equation.

Numerical calculation. The background and exponents are

$$\omega_m = \frac{m}{2} \mathbf{1}_{[-1,1]}, \quad (a, r, m) = (0.8, 0.2, 1.8).$$

For a trial support $[-L, L]$, partition the interval into n uniform cells $[e_{j-1}, e_j]$ of width $h = 2L/n$, with midpoints x_j . Write $\phi_h = \sum_j c_j \mathbf{1}_{[e_{j-1}, e_j]}$. Midpoint collocation gives

$$Ac = b, \quad A_{ij} = r(1+r) \int_{e_{j-1}}^{e_j} |x_i - y|^{r-1} dy, \\ b_i = \frac{m(1+a)}{2} [\operatorname{sgn}(x_i + 1)|x_i + 1|^a - \operatorname{sgn}(x_i - 1)|x_i - 1|^a].$$

Each cell integral, including the diagonal singular cell, is evaluated analytically:

$$\int_{\ell}^u |x-y|^{r-1} dy = \begin{cases} ((x-\ell)^r - (x-u)^r)/r, & x \geq u, \\ ((u-x)^r - (\ell-x)^r)/r, & x \leq \ell, \\ ((x-\ell)^r + (u-x)^r)/r, & \ell < x < u. \end{cases}$$

What are A , b , and c ? The continuum equation is $\mathcal{T}_L \phi = G_a$, where

$$(\mathcal{T}_L \phi)(x) = r(1+r) \int_{-L}^L |x-y|^{r-1} \phi(y) dy.$$

The vector $c = (c_1, \dots, c_n)^\top$ contains the unknown *density values* on the cells; the mass of cell j is hc_j . The matrix $A \in \mathbb{R}^{n \times n}$ is the discretized operator $\mathcal{T}_L$: its entry A_{ij} is the contribution of a

unit density on cell j to the equation evaluated at x_i . The vector $b = (b_1, \dots, b_n)^\top$ contains the known values $b_i = G_a(x_i) = V''(x_i)$ generated by the background. In particular,

$$(Ac)_i = \sum_{j=1}^n c_j r (1+r) \int_{e_{j-1}}^{e_j} |x_i - y|^{r-1} dy = (\mathcal{T}_L \phi_h)(x_i).$$

Thus $Ac = b$ imposes the original integral equation on the piecewise-constant approximation at every midpoint. Here A is a matrix; the endpoint notation $I = [A, B]$ used in Section 2.2 is local to that section and denotes scalar interval endpoints.

MATLAB's backslash operator solves the linear system. The support is selected by `fzero` applied to $M_{140}(L) - 1$ on $[0.16, 0.45]$, where $M_n(L) = h \sum_j c_j$. No custom solver tolerances are specified. At the selected L , the system is recomputed with $n = 320$; the support root is not recomputed on this finer grid. The script prints its mass and relative algebraic residual

$$\mathcal{R}_{\text{Riesz}} = \|Ac - b\|_2 / \|b\|_2.$$

The density curve uses the raw coefficients c_j , without clipping. The cumulative distribution is constructed at cell edges by integration and divided by its final mass. Thus CDF normalization is distinct from the reported mass of the computed density.

Meaning of the output. The result is a stationary reference density and its cumulative mass, together with a computed support half-width. The linear residual measures how accurately the discrete integral equation is solved; it does not by itself measure quadrature error or prove positivity of the continuum solution. These outputs form the finite-Riesz density and CDF curves in the one-dimensional figure.

2.2 Green/Boggio reconstruction of the interval profile (Proposition 7.1)

Original problem. The target is the second stationary curve in the one-dimensional figure, called the Green-kernel reconstruction in Proposition 7.1 and the figure caption. For $I = [-L, L]$, $\Omega = \mathbb{R} \setminus I$, and $\sigma = r/2$, the proposition gives

$$H(x) = c_{1,r}^{-1} \int_{\Omega} \mathcal{G}_{\Omega}^{\sigma}(x, y) Q_{a,r}(y) dy, \quad x \in \Omega, \quad H = 0 \quad \text{on } I,$$

$$\phi_I = (Q_{a,r} - c_{1,r}(-\Delta)^{\sigma} H)|_I.$$

Here $\mathcal{G}_{\Omega}^{\sigma}$ is the exterior Green kernel (the Green/Boggio representation in the paper), $Q_{a,r} = \mathcal{L}_r V$, and $c_{1,r}$ is the normalization in $\mathcal{L}_r = c_{1,r}(-\Delta)^{1+\sigma}$.

Equivalence of the Boggio/Green and Poisson formulations. These are two equivalent formulations of the same continuum construction, with different unknowns. The equivalence requires the corresponding solution spaces and behavior at infinity; it does not assert equality of their finite-resolution approximations. Set

$$k = r(1+r), \quad g = V''/k, \quad U(x) = \int_I |x - z|^{r-1} \phi(z) dz.$$

For the algebra below, $g = V''/k$ is defined on the whole real line, although the Poisson integral uses only its values on I . The manuscript uses $F = V - \psi_r * \phi$ and $H = (-\Delta)F$. Since $(\psi_r * \phi)'' = kU$, these definitions give the exact change of unknown

$$H = kU - V'' = k(U - g), \quad U = g + H/k.$$

In particular, H and U are not the same function. With $D_\sigma = (-\Delta)^\sigma$, the source identity is

$$Q_{a,r} = c_{1,r}(-\Delta)^{1+\sigma}V = -c_{1,r}D_\sigma V'' = -c_{1,r}kD_\sigma g.$$

The two boundary-value problems are therefore

$$\underbrace{\begin{cases} D_\sigma H = c_{1,r}^{-1}Q_{a,r} = -kD_\sigma g & \text{in } \Omega, \\ H = 0 & \text{on } I \end{cases}}_{\text{Boggio/Green formulation for } H} \iff \underbrace{\begin{cases} D_\sigma U = 0 & \text{in } \Omega, \\ U = g & \text{on } I \end{cases}}_{\text{Poisson formulation for } U}.$$

Indeed, outside I , $D_\sigma U = D_\sigma g + D_\sigma H/k = 0$; on I , $H = 0$ gives $U = g$. Conversely, $H = k(U - g)$ recovers both equations for H . The source is absorbed into the change of unknown; it has not been discarded. For the compactly supported finite-mass potential considered here, $U(x) = O(|x|^{r-1})$ at infinity. The corresponding condition on H is imposed through $H = kU - V''$. The representations are compared in these matched classes, with the well-posedness and operator assumptions of the manuscript.

How the two integral formulas represent the same solution. Write $P_\Omega^\sigma(y, x)$ for the exterior Poisson kernel, with $y \in \Omega$ and $x \in I$. Substitution of the Boggio/Green representation of H into $U = (H + V'')/k$ gives

$$\begin{aligned} U(y) &= \frac{1}{k} \left[V''(y) + c_{1,r}^{-1} \int_\Omega \mathcal{G}_\Omega^\sigma(y, z) Q_{a,r}(z) dz \right] \\ &= \int_I P_\Omega^\sigma(y, x) \frac{V''(x)}{k} dx. \end{aligned}$$

The first line obtains U from the manuscript's Boggio solution; the second is its Poisson representation. Equivalently,

$$H(y) = \int_I P_\Omega^\sigma(y, x) V''(x) dx - V''(y).$$

Thus the two kernels have different arguments and roles, but the two formulas are related by an explicit change of unknown. Calling the code's kernel a Poisson kernel does not introduce a different stationary problem.

Why the recovered density is the same. Substituting $H = kU - V''$ into the manuscript's global reconstruction gives

$$\begin{aligned} \phi_I &= Q_{a,r} - c_{1,r}D_\sigma H \\ &= Q_{a,r} - c_{1,r}kD_\sigma U + c_{1,r}D_\sigma V'' \\ &= -c_{1,r}kD_\sigma U. \end{aligned}$$

The source and the V'' term cancel exactly. Under the manuscript's inverse-kernel normalization, $-c_{1,r}k$ is the positive Riesz-inversion factor. This explains the two operations in `formula516Density1D`: first compute $U = g$ on I and its Poisson extension outside; then apply the singular-integral formula for $D_\sigma U$ at interior nodes. Exterior values are needed because this operator is nonlocal. The omitted positive common factor is removed by the code's final mass normalization. Numerical truncation, clipping and normalization remain separate approximation steps, described below.

Why the code uses Poisson in one dimension and Boggio in two. This is a choice in the supplied implementation, not a dimension-dependent mathematical requirement. The one-dimensional routine uses the explicit interval Poisson kernel to evaluate the equivalent potential U . The two-dimensional routine in Section 3.2 instead evaluates the exterior Boggio/Green integral for H_R directly and then reconstructs the density. Exterior disks also admit Poisson representations; the code does not implement that alternative. No benchmark in the supplied script establishes that either choice is faster or more accurate. Moreover, the two-dimensional routine fits a relative coefficient, as explained in Section 3.2, so the continuum equivalence above must not be read as a claim of exact agreement between the implemented numerical curves.

Numerical calculation. This calculation uses the prescribed half-width $L_P = 0.2727$, $A = -L_P$, $B = L_P$, $n = 320$ interior cells, and $\sigma = r/2 = 0.1$. At interior midpoints set $g_i = b_i/[r(1+r)]$. For $y \notin [A, B]$,

$$\tilde{g}_h(y) = h \sum_i P(y, x_i) g_i, \quad P(y, x) = \frac{\sin(\pi\sigma)}{\pi} \left[\frac{(y-A)(y-B)}{(x-A)(B-x)} \right]^\sigma \frac{1}{|y-x|}.$$

The denominator $(x-A)(B-x)$ is floored at MATLAB's `eps`. The exterior-domain kernel above is the one implemented in this version.

On each side, the exterior integral uses $n_o = 1400$ nodes, $Y_{\max} = 10$, and grading exponent $\beta = 1.8$:

$$u_j = \frac{j-1/2}{n_o}, \quad z_j = (Y_{\max} - L_P) u_j^\beta + 10^{-6}, \quad w_j = \frac{(Y_{\max} - L_P)^\beta}{n_o} u_j^{\beta-1},$$

with $y_j^- = A - z_j$ and $y_j^+ = B + z_j$. Up to a common multiplicative constant, the reconstructed values are

$$\begin{aligned} f_i = h \sum_{\ell \neq i} \frac{g_i - g_\ell}{|x_i - x_\ell|^{1+2\sigma}} \\ + \sum_j \frac{g_i - \tilde{g}_h(y_j^-)}{|x_i - y_j^-|^{1+2\sigma}} w_j + \sum_j \frac{g_i - \tilde{g}_h(y_j^+)}{|x_i - y_j^+|^{1+2\sigma}} w_j. \end{aligned}$$

Interior terms at distance at most 10^{-12} are excluded. No correction for the omitted exterior tail is added. The code computes $M_- = h \sum_i \max(-f_i, 0)$ and prints a diagnostic if $M_- > 10^{-4}$. It then clips and normalizes:

$$\hat{f}_i = \frac{\max(f_i, 0)}{h \sum_\ell \max(f_\ell, 0)}.$$

If the denominator is nonpositive, a warning is issued and the normalized nonnegative part of g is used instead. The omitted common multiplicative constant cancels in this normalization. The resulting CDF is normalized once more by its final cumulative mass.

The finite-Riesz CDF is linearly interpolated to the Green/Boggio comparison curve's cell edges, with linear extrapolation if necessary. The reported discrepancy is the maximum absolute difference on those edges. It compares normalized CDFs on two potentially different support grids.

Meaning of the output. This route produces the second stationary CDF in the one-dimensional figure. Comparing it with the finite-Riesz CDF checks agreement of two numerical realizations of the interval characterization. The fixed, rounded half-width, finite exterior cutoff, endpoint quadrature, clipping, and normalization all enter this comparison. Because a common constant is removed by normalization, the displayed CDF comparison tests profile shape rather than the unnormalized reconstruction mass.

2.3 Particle dynamics and diagnostics

Original problem and purpose. The evolution equation transports a probability measure with velocity $-K_a * \omega_m + K_r * \phi_t$. Replacing that measure by N equally weighted point masses yields the particle ODE below. Starting away from the computed stationary profile and comparing its distribution at $T = 50$ with Sections 2.1–2.2 tests whether this trajectory approaches the predicted profile in the displayed experiment.

Numerical calculation. The particle system has $N = 500$, time step $\Delta t = 0.02$, and 2500 explicit Euler steps, giving $T = 50$. With `rng(12)`, its initial positions are

$$X_i^0 = -1.4 + \frac{2.8(i-1)}{N-1} + 0.01\xi_i,$$

where ξ_i are the values returned by `randn`. Initial positions are sorted. The exact background force is

$$F_\omega(x) = \frac{m}{2} (|x+1|^{1+a} - |x-1|^{1+a}),$$

and the update is

$$X_i^{k+1} = X_i^k + \Delta t \left[-F_\omega(X_i^k) + \frac{1+r}{N} \sum_j \text{sgn}(X_i^k - X_j^k) |X_i^k - X_j^k|^r \right].$$

All pairwise terms use the old positions. The diagonal is explicitly set to zero. No force smoothing is used. Positions are sorted every 20 steps and after the final step. Termination is at the fixed time T , with no residual-based early stopping.

Every four steps the script records the diameter $D = \max_i X_i - \min_i X_i$ and twice the MATLAB 99% quantile of $|X_i|$. These are plotted with $2L$. For the empirical CDF $F_N(x, T) = N^{-1} \sum_i \mathbf{1}_{\{X_i(T) \leq x\}}$ and normalized Riesz CDF F_h , the reported error is

$$\mathcal{E}_{\text{CDF}} = \max_{0 \leq j \leq 320} |F_N(e_j, T) - F_h(e_j)|.$$

This is a maximum on the selected cell edges, not an independently computed supremum over all real x . The histogram uses 55 equally spaced edges (54 bins) on $[-1.05L, 1.05L]$ and `Normalization='pdf'`.

2.4 Reading the one-dimensional figure

Left picture: density and particle histogram. The horizontal axis is position x and the vertical axis is probability density. The solid curve is the finite-Riesz density from Section 2.1; the dashed histogram is formed from the evolved particles at $T = 50$. The vertical dotted lines mark the predicted endpoints $-L$ and L , with $L \approx 0.2727$. In the supplied figure the reference density rises near these endpoints and is lower near the center. The histogram follows this shape, with bin-to-bin fluctuations expected for finitely many particles. This panel compares the spatial density profile, not the background density. Histogram binning can obscure endpoint behavior, so the middle picture also compares cumulative masses.

Middle picture: three cumulative distributions. At position x , each curve gives the probability mass to the left of x . The three curves are the integrated finite-Riesz density, the normalized Green/Boggio reconstruction from Section 2.2 at $L = 0.2727$, and the empirical CDF of the particles at $T = 50$. Their close overlap in the supplied figure shows agreement of the two stationary-profile computations with the evolved particle distribution at the displayed resolution. Values near zero to the left and near one to the right identify the support transition. The figure title reports a maximum CDF discrepancy of 1.09×10^{-3} ; this is the particle-versus-finite-Riesz discrepancy on the selected cell edges defined in Section 2.3, not the error of every curve against an exact solution.

Right picture: evolution of the spatial extent. The horizontal axis is time. The solid trace is the full particle diameter, and the dashed trace is twice the 99% quantile of the absolute particle positions. The horizontal dotted line is the stationary support length $2L \approx 0.5454$. The supplied figure shows both traces decreasing rapidly from their initial values and then remaining close to the predicted length. The full diameter depends on the outermost particles, while the quantile trace is less sensitive to a small number of extreme positions. This panel tests support-size agreement; it complements the density and CDF comparisons rather than determining the entire profile.

3 Two-dimensional radial experiment

3.1 Solving the two-dimensional stationary radial force balance

Original problem. For the Gaussian background, seek a radial stationary probability measure supported in a disk of radius R . The mathematical starting point is the radial form of the zero-force condition, stated explicitly below. Write the radial mass measure as

$$d\mu(\eta) = 2\pi\eta\phi(\eta) d\eta$$

when it has a density. The force condition and mass constraint are

$$\int_0^R \mathcal{A}_r(\rho, \eta) d\mu(\eta) = m \int_0^\infty \mathcal{A}_a(\rho, \eta) \eta e^{-\eta^2/2} d\eta, \quad 0 < \rho < R, \quad \mu([0, R]) = 1.$$

The two kernels in this equation are explicitly

$$\begin{aligned} \mathcal{A}_r(\rho, \eta) &= \frac{1+r}{2\pi} \int_0^{2\pi} (\rho - \eta \cos \theta)(\rho^2 + \eta^2 - 2\rho\eta \cos \theta)^{(r-1)/2} d\theta, \\ \mathcal{A}_a(\rho, \eta) &= \frac{1+a}{2\pi} \int_0^{2\pi} (\rho - \eta \cos \theta)(\rho^2 + \eta^2 - 2\rho\eta \cos \theta)^{(a-1)/2} d\theta. \end{aligned}$$

To derive these expressions, place the observation point at $x = (\rho, 0)$ and a source point at $y = (\eta \cos \theta, \eta \sin \theta)$. For $K_\alpha(z) = (1 + \alpha)z|z|^{\alpha-1}$, its radial component is

$$K_\alpha(x - y) \cdot (1, 0) = (1 + \alpha)(\rho - \eta \cos \theta)(\rho^2 + \eta^2 - 2\rho\eta \cos \theta)^{(\alpha-1)/2}.$$

Averaging over the angle gives $\mathcal{A}_\alpha$; tangential components cancel by symmetry. The kernel $\mathcal{A}_r$ acts on the unknown stationary radial mass, whereas $\mathcal{A}_a$ acts on the known Gaussian background radial mass $m\eta e^{-\eta^2/2} d\eta$.

Why solve for ring masses? Each unknown is a mass distributed uniformly around a circle, rather than a density value at a Cartesian point. Its force contribution is its mass times $\mathcal{A}_r$. Thus the force equation is linear in the unknown ring masses. Nonnegative least squares imposes nonnegativity, while an appended weighted equation encourages unit total mass. Using more test radii than unknown rings makes the force system overdetermined and provides sampled residuals. The supplied code prescribes R ; it does not solve a free-boundary optimization over R in this calculation.

Numerical calculation. Here

$$(a, r, m) = (0.6, 0.2, 1.8), \quad \omega_m(x) = \frac{m}{2\pi} e^{-|x|^2/2}, \quad R = 0.33.$$

The radius R is fixed in the script. A source comment attributes this choice to a coarse residual scan, but that scan is not implemented in the supplied file. The radial force of a unit circular mass is approximated as

$$\mathcal{A}_{\alpha,q}(\rho, s) = \frac{1+\alpha}{q} \sum_{\ell=1}^q (\rho - s \cos \theta_\ell) [\max(\rho^2 + s^2 - 2\rho s \cos \theta_\ell, 10^{-14})]^{(\alpha-1)/2},$$

where $\theta_\ell = 2\pi(\ell - 1/2)/q$. The distance floor is applied before evaluating the power.

The stationary discretization uses 110 source rings at $s_j = (j - 1/2)R/110$, 150 test radii $\rho_i = (i - 1/2)R/150$, and $q = 160$ angular nodes. The background uses 550 radial midpoint nodes on $[0, 6]$:

$$\Delta s_b = 6/550, \quad s_\ell^b = (\ell - 1/2)\Delta s_b, \quad w_\ell^b = m s_\ell^b e^{-(s_\ell^b)^2/2} \Delta s_b.$$

These truncated weights are not renormalized. Set

$$A_{ij} = \mathcal{A}_{r,160}(\rho_i, s_j), \quad b_i = \sum_{\ell} \mathcal{A}_{a,160}(\rho_i, s_\ell^b) w_\ell^b.$$

The code solves

$$\min_{\mu \geq 0} \{ \|A\mu - b\|_2^2 + 120^2 (\mathbf{1}^\top \mu - 1)^2 \}$$

with `lsqnonneg`, using default settings. This is midpoint force collocation with a nonnegative least-squares solve (called the radial Galerkin curve in the script). The result is clipped and divided by its total mass.

If `lsqnonneg` is unavailable, define $\tilde{A} = [A; 120\mathbf{1}^\top]$ and $\tilde{b} = [b; 120]$. Starting with the nonnegative part of the backslash solution, perform 800 iterations

$$\mu^{k+1} = \max\{\mu^k - \tau \tilde{A}^\top (\tilde{A}\mu^k - \tilde{b}), 0\}, \quad \tau = 0.9/(\|\tilde{A}\|_2^2 + 10^{-12}).$$

The final normalized solution defines $\text{Res} = A\mu - b$. The plot shows $|\text{Res}_i|$, and the console reports $\|\text{Res}\|_2/\sqrt{150}$ and $\|\text{Res}\|_\infty$. These are sampled stationary force residuals, not dynamical stopping criteria.

Meaning of the output. The normalized ring masses define the first stationary radial CDF. The plotted residual measures the remaining force imbalance at the test radii after normalization. This calculation solves the force-verification equation directly and does not evaluate the exterior Green formula.

3.2 Constructing a disk profile through the exterior Green formula

Original problem. The second route constructs a candidate density on the disk by solving an auxiliary fractional equation outside it. Here $Q_{a,r}$ is the source obtained by applying the inverse repulsive-potential operator to $\psi_a * \omega_m$, and $\lambda_{2,r}$ is its normalization constant. For the Gaussian background, the source is evaluated through the Hankel integral given below. Set $s = (1 + r)/2$ and solve

$$(-\Delta)^s H_R = \lambda_{2,r}^{-1} Q_{a,r} \quad \text{in } B_R^c, \quad H_R = 0 \quad \text{in } B_R.$$

Its Green representation gives H_R on the exterior. The candidate density on the disk is then

$$\phi_R = (Q_{a,r} - \lambda_{2,r}(-\Delta)^s H_R) \big|_{B_R}.$$

As stated in the proposition, the candidate must also satisfy nonnegativity, mass one, and the force condition to represent a zero-flux stationary probability measure.

Why are there two exterior integrations? First the exterior Green integral maps the source $Q_{a,r}$ to H_R . Next, because $H_R(x) = 0$ for x inside the disk, the fractional Laplacian there reduces to

$$(-\Delta)^s H_R(x) = -\kappa_{2,s} \int_{B_R^c} \frac{H_R(y)}{|x-y|^{3+r}} dy.$$

The second integration therefore maps the exterior potential back to a bulk correction inside the disk. The reconstructed profile consists of the source term plus this correction, with the prescribed theoretical constants. For the Gaussian background, radial Fourier inversion expresses the source shape as a Hankel integral involving J_0 . These three steps explain the code's source calculation, Green calculation, and interior correction calculation.

Numerical calculation and its calibration. The supplied implementation calculates shapes with some normalization constants suppressed and then fits a relative coefficient using the force residual from Section 3.1. The following formulas describe that implemented procedure, rather than claiming that all theoretical constants have been evaluated independently. Let $s = (1+r)/2 = 0.6$. The source shape is evaluated using 2500 uniformly spaced frequencies k_ℓ from 10^{-5} to 16:

$$q_h(\rho) = \max \left\{ m \Delta k \sum_{\ell} k_{\ell}^{1+r-a} e^{-k_{\ell}^2/2} J_0(k_{\ell} \rho), 0 \right\}.$$

The computation suppresses overall Fourier normalization constants. Exterior quadrature uses 85 nodes with grading exponent 1.35:

$$u_j = (j - 1/2)/85, \quad \eta_j = R + (4 - R)u_j^{1.35}, \quad w_j = (4 - R)1.35u_j^{0.35}/85.$$

Angular averages use 96 midpoint nodes. For exterior ρ, η , define

$$d_{\ell}^2 = \max\{\rho^2 + \eta^2 - 2\rho\eta \cos \theta_{\ell}, 10^{-12}\}, \quad z_{\ell} = \frac{(\rho^2 - R^2)(\eta^2 - R^2)}{R^2 d_{\ell}^2}.$$

The kernel shape is

$$G_h(\rho, \eta) = \frac{1}{96} \sum_{\ell} (d_{\ell}^2)^{-(1-r)/2} I_s(z_{\ell}), \quad I_s(z) = \frac{z^s}{s} {}_2F_1(s, 1; s+1; -z).$$

The auxiliary function $I_s(z) = \int_0^z t^{s-1}/(1+t) dt$ is evaluated by `hypergeom`, with negative arguments z clipped to zero before evaluation. Set

$$H_i = \sum_j G_h(\eta_i, \eta_j) q_h(\eta_j) 2\pi \eta_j w_j.$$

For interior radii s_i , the correction is

$$C_i = \sum_j H_j 2\pi \eta_j w_j \frac{1}{96} \sum_{\ell} [\max\{s_i^2 + \eta_j^2 - 2s_i \eta_j \cos \theta_{\ell}, 10^{-12}\}]^{-(3+r)/2}.$$

No tail correction beyond exterior radius 4 is included. With $\Delta s = R/110$, form

$$\nu_i(c) = \frac{\max\{q_h(s_i) + cC_i, 0\} 2\pi s_i \Delta s}{\sum_j \max\{q_h(s_j) + cC_j, 0\} 2\pi s_j \Delta s}.$$

The coefficient is selected by `fminbnd`, minimizing $\|A\nu(e^{\ell}) - b\|_2/\sqrt{150}$ for $\ell \in [-10, 10]$, with default settings. If the optimization throws an error, $c = 1$ is used. In the objective function only, nonpositive total mass triggers a uniform-mass fallback.

This coefficient changes the relative contributions of q_h and C ; it is not simply an overall normalization. Thus the displayed reconstruction includes a scalar fit to the same force equation used for the stationary ring masses. A fully independent evaluation with all constants fixed analytically would require a different implementation.

Meaning of the output. The output is the second stationary radial CDF. It is assembled through exterior quadratures instead of the ring-mass least-squares solve. However, its scalar calibration uses the same force matrix and background vector as Section 3.1, so the two curves are not fully independent numerical checks. Their comparison measures agreement of the implemented constructions, including that calibration.

3.3 Radial evolution and visualization

Original problem and purpose. For a radial solution, the characteristic radius satisfies the background attraction plus the self-repulsion averaged over angles. Approximating the evolving radial mass by equal-weight circular masses yields the ODE discretized below. Its final radial CDF is compared with the stationary CDFs from Sections 3.1–3.2. The Cartesian cloud is a visualization of those final radii.

Numerical calculation. The dynamics uses $N = 180$ equal-weight radial nodes, initialized deterministically by

$$\rho_i^0 = 1.1\sqrt{-2\log(1 - 0.98u_i)}, \quad u_i = (i - 1/2)/N.$$

The force tables use 260 uniform radii on $[0, 3.2]$ and 96 angular nodes. The background table uses the same 550 background nodes. Background queries use linear interpolation and linear extrapolation; self-force queries use bilinear interpolation with zero outside the table domain. Denote these interpolants by $\tilde{F}_\omega$ and $\tilde{\mathcal{A}}_r$. The explicit Euler update is

$$\rho_i^{k+1} = \max \left\{ \rho_i^k + \Delta t \left[-\tilde{F}_\omega(\rho_i^k) + \frac{1}{N} \sum_j \tilde{\mathcal{A}}_r(\rho_i^k, \rho_j^k) \right], 0 \right\}.$$

The diagonal ring contribution is retained. A radial node represents mass uniformly distributed over a circle, so this contribution differs from a point-particle diagonal force. Radii are sorted after every step. The time step is 0.05, and 1000 steps give $T = 50$. There is no residual-based early stopping.

Stationary cumulative masses $\sum_{s_j \leq \rho} \mu_j$ and $\sum_{s_j \leq \rho} \nu_j$ are evaluated on 500 radii in $[0, 0.66]$ and plotted against the empirical radial CDF $N^{-1} \sum_i \mathbf{1}_{\{\rho_i(T) \leq \rho\}}$. The script does not print a two-dimensional CDF error. The cloud visualization assigns angles

$$\vartheta_i = (i - 1)\pi(3 - \sqrt{5}) \pmod{2\pi}, \quad Z_i = \rho_i(T)(\cos \vartheta_i, \sin \vartheta_i).$$

These angles are for display only; Cartesian particle trajectories are not integrated.

3.4 Reading the two-dimensional figure

Left picture: final radial distribution in the plane. The axes are the Cartesian coordinates x_1, x_2 . Each marker is placed at an evolved radius and a deterministic display angle, as described in Section 3.3. The dashed circle marks the candidate support boundary $R = 0.33$. In the supplied figure the points occupy the disk and reach close to this circle. This visualizes the radial distribution at $T = 50$; the angular arrangement is assigned for display and does not show independently evolved angular motion. The picture is a qualitative support check, while the middle picture quantifies radial mass allocation.

Middle picture: three cumulative radial masses. The horizontal axis is radius ρ , and the vertical axis is the total probability inside the centered disk B_ρ . The solid curve is obtained from the zero-flux ring-mass solve in Section 3.1 (called the Galerkin curve in the legend). The second

stationary curve comes from the exterior Green reconstruction of Section 3.2, and the particle curve is the empirical radial mass at $T = 50$. The vertical dotted line marks $R = 0.33$. In the supplied figure the curves nearly overlap, rise steeply near the support boundary, and attain mass one near R . This indicates agreement in how mass is distributed by radius, including substantial mass near the outer part of the disk. Recall that the Green reconstruction's scalar coefficient is calibrated against the same force equation; this dependence is part of the implementation.

Right picture: stationary force residual by radius. The horizontal axis is the test radius and the vertical axis is the absolute residual $|(A\mu - b)_i|$ from Section 3.1, evaluated after mass normalization. It measures the mismatch between the computed repulsive force and the background attraction at each test radius. The supplied plot shows small residuals over most of the disk and a sharp increase near its outer boundary; the displayed vertical scale is 10^{-5} . This identifies where the sampled force balance is least accurately satisfied. The curve belongs to the zero-flux stationary ring solution, not to the particle evolution and not to the separately reconstructed Green curve. It is a force residual, not a density error or a time-dependent convergence rate.